FlowConX

Risk-controlled throughput forecasting and bandwidth allocation for Starlink (LEO satellite) access links

Mayan Sharma (23BAI1581) Kriti Saini (23BCE1520) Devansh Behl (23BCE1334)

Guide: Dr. Suganya R, Professor, SCOPE

Industry mentor: Mr. S. Vijayaragavan, Professor of Practice, Nelco Network Products Ltd.

School of Computer Science and Engineering, Vellore Institute of Technology

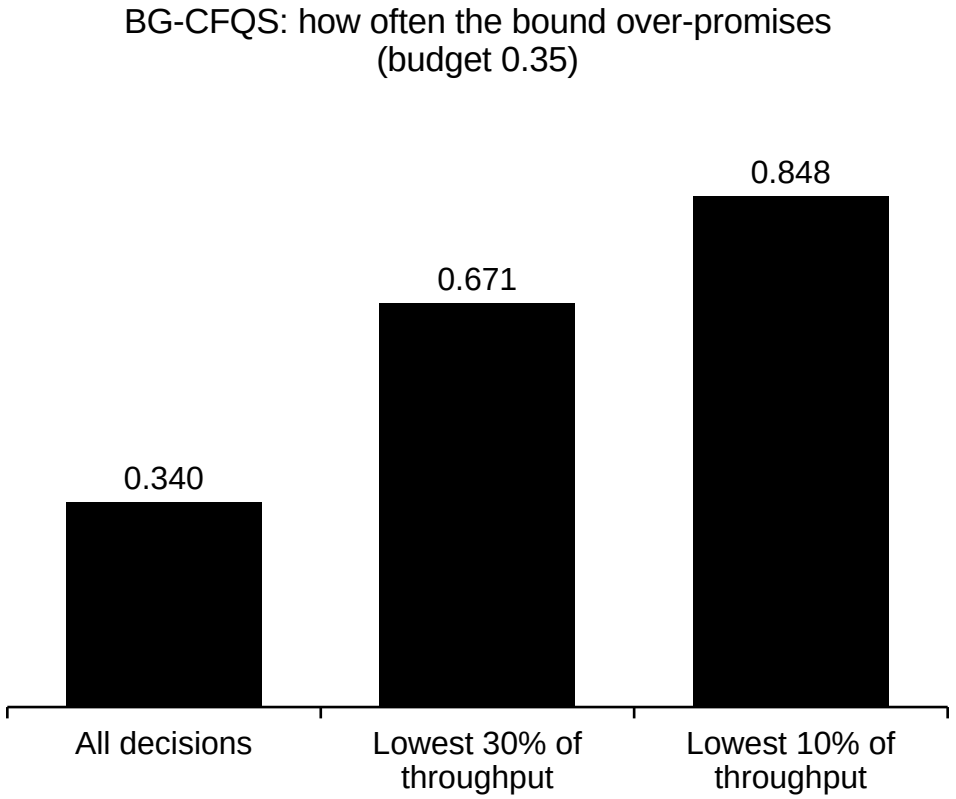
Domain Knowledge and Problem Understanding

The problem

- Starlink capacity changes every 15 s: the network reschedules users at seconds 12, 27, 42 and 57 of each minute, and satellites hand over constantly.
- An operator must commit bandwidth before it knows what the link will carry. Over-promise and sessions drop; under-promise and paid capacity sits idle.
- Industry brief: predict degradation, detect congestion early, allocate bandwidth automatically. We chain four needs into one system, adding: decide which flow is slowed when capacity is short.

Prior work

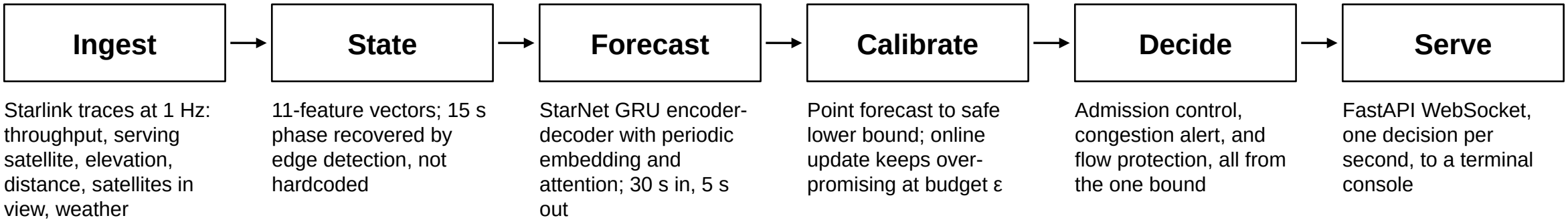
- StarNet (Liu et al., CoNEXT 2025): point forecast of throughput.
- BG-CFQS (Xie et al., 2026): safe lower bound from one global quantile.
- Casparsen et al. (2026): the 15 s scheduling period. Horizon (SIGMETRICS 2026): global Starlink performance.



Our reproduction of BG-CFQS. The risk is held on average and lost exactly where capacity is lowest, which is where over-allocation drops sessions.

Methodology, Architecture and Technical Design

Data flow: each layer only talks to the one below it, so each can be tested on its own.



Algorithms

- Online conformal calibration: after each outcome, $\alpha \leftarrow \alpha + \gamma(\varepsilon - \text{err})$, so the bound tracks the budget as the link drifts.
- Admission: floor(bound / 10 Mbps) sessions. Congestion: bound below the committed rate for 5 decisions in a row.
- Flow protection: criticality scored online from 7 signals (attention, interactivity, elasticity, deadline, irreversibility, volume, recurrence), then the bound is split by weighted max-min fairness with protected minimums.

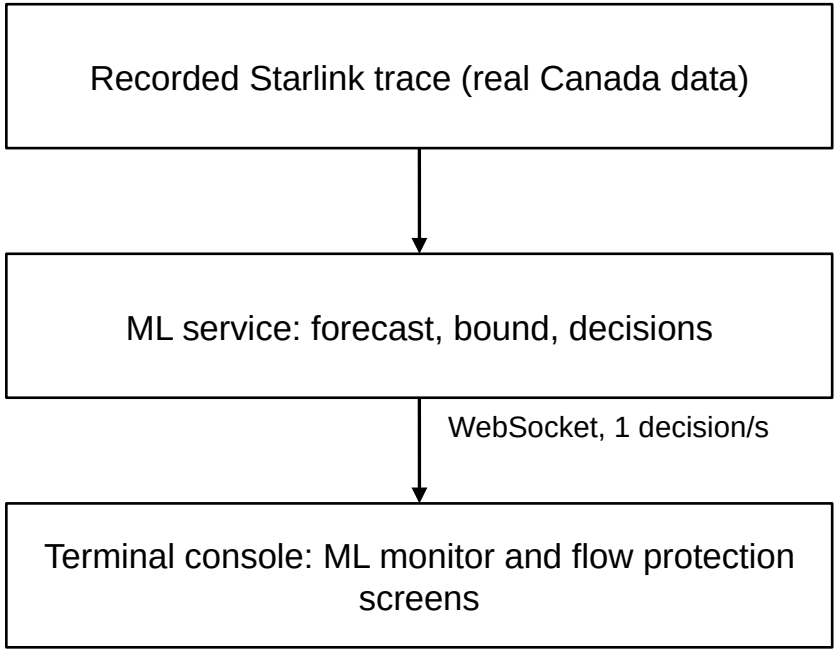
Technical decisions

- Temporal train / calibration / test splits. Random splits flatter these models.
- The calibration set is used only to set the operating point, never to fit.
- Congestion is derived from the bound, not a separate classifier.
- No flow is ever cut to zero while capacity remains.
- Every run saves its configuration and seed.

Completion, Functionality and Integration

Module	What it delivers	Status
Throughput forecast	StarNet reproduced on USA, Canada, Germany	Complete
Safe bound	Online calibration at any risk budget	Complete
Admission control	Sessions admitted from the safe bound	Complete
Congestion alert	Early warning from the same bound	Complete
Flow protection	Decides which flow is slowed, never stopped	Complete
Latency, availability, cost	Measured beyond the approved scope	Complete
Live system	ML service and terminal console on real traces	Complete

Integration at run time



- Runs end to end on a laptop; the replay loops continuously without restarts.
- 328 automated tests pass on every change.

Testing, Validation and Analysis of Results (1/2)

How it was tested

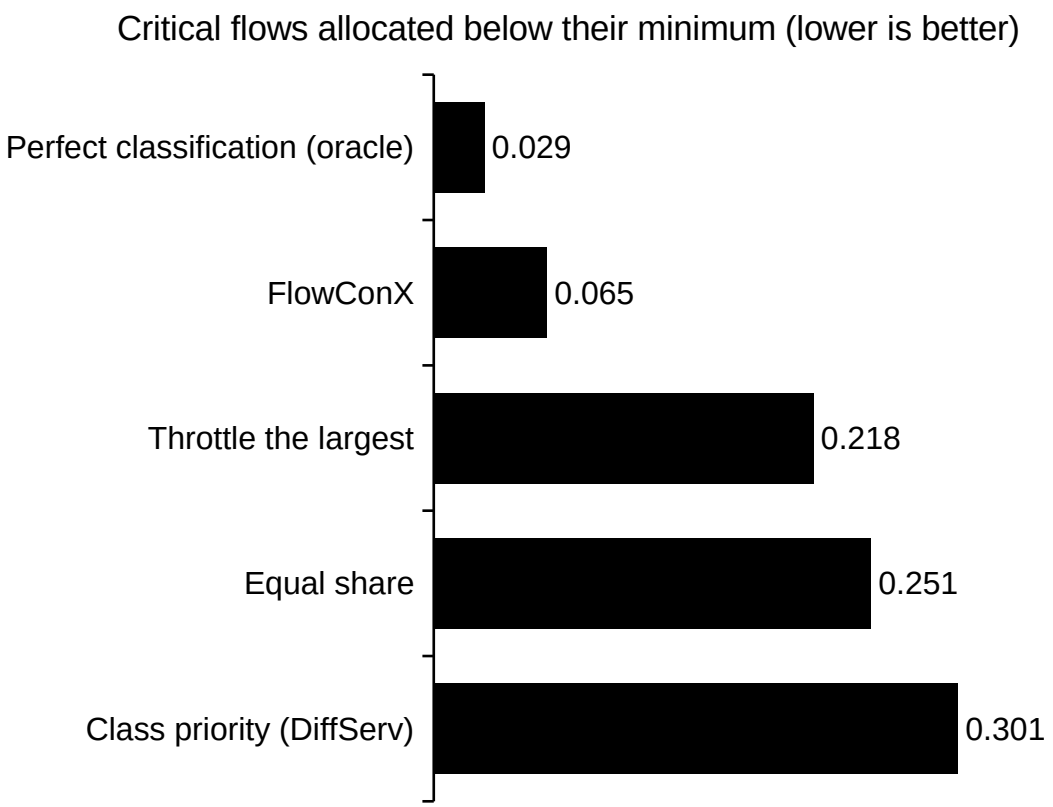
- Reproduce the published baselines first; no new method until both match.
- Temporal splits and leave-one-location-out across USA, Canada and Germany, plus a second measurement campaign (WetLinks).
- Data loader checked against published sample counts: 145,053 (Canada), 613,295 (Germany), 1,123,832 (USA), all exact.
- 328 automated tests, including checks that no decision uses future data.
- Every number is read from a saved result with its configuration and seed.

Reproduction against published results

Check	Published	FlowConX
StarNet RMSE, average (Mbps)	39.30	39.73 (+1.1%)
StarNet MAE, average (Mbps)	29.28	29.97 (+2.4%)
BG-CFQS over-promise rate	0.349	0.340
BG-CFQS, lowest 30%	0.65 to 0.71	0.671
BG-CFQS, lowest 10%	0.83 to 0.86	0.848
15 s phase offset (s)	12	11.98 / 12.09 / 12.25

Both reproduction checks passed. Per-location StarNet gaps scatter in both directions (4.2% to 6.9%), as an independent reimplementation should.

Testing, Validation and Analysis of Results (2/2)



Canada link, 2x load, 5 workload seeds. Critical transfers that gave up: 0.3% with FlowConX against 12.1% when the largest flow is throttled first. Capacity is measured; flows are simulated.

Key results

Measure	Before	FlowConX
Risk budget held (throughput)	16% to 21% off	within 1%
Availability	about 1.2 nines	about 2.1 nines
Number of outages	baseline	5 to 6x fewer
Latency budget 0.10 held	0.033 to 0.313	0.088 to 0.101

What did not work, reported as found

- Conditioning on satellite geometry did not improve risk control on StarNet.
- Congestion alerts fire early but are noisy (F1 0.46).
- The scorer cannot tell a video call from a video stream (precision 0.71).
- BG-CFQS cannot serve budgets below 0.15 (3.7x overshoot at 0.05).

Innovation, Technical Merit and Practical Relevance

Contribution

- First evaluation of risk-controlled capacity forecasting on LEO links: 4 datasets, 2 measurement campaigns.
- Showed BG-CFQS needs exchangeable data and has a budget floor.
- Flow protection splits the safe bound, so a reserved minimum is a promise that can be checked.

Feasibility

- Runs on a laptop: CPU, CUDA or Apple MPS.
- The demo model trains in about 75 s and decides once per second.
- Uses only data a terminal has at prediction time.

Ethical responsibility

- Public measurement traces; no personal data.
- Simulated flows are labelled as simulated on screen.
- No flow is stopped while capacity remains; self-declared priority is not trusted blindly.
- Failures and withdrawn claims are reported.

Industry validation

- Demonstrated live to Mr. S. Vijayaragavan, Professor of Practice, Nelco Network Products Ltd., who was very impressed with the system.
- He set the problem statement and co-authored our paper with us, prepared for submission to ACM SIGMETRICS 2027.
- Practical use: satellite operators can commit bandwidth at a stated risk and keep critical traffic, such as a hospital upload, running when the link is short.